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A randomised controlled phase 3 trial of lutetium-177-PSMA-617 in taxane-naïve patients with progressive metastatic castration-resistant prostate cancer (PSMAfore)

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Summary

BACKGROUND: [^{177}Lu]Lu-PSMA-617 (^{177}Lu -PSMA-617) prolongs radiographic progression-free survival (rPFS) and overall survival (OS) in patients with metastatic castration-resistant prostate cancer (mCRPC) previously treated with androgen receptor pathway inhibitor (ARPI) and taxane therapy. PSMAfore is an ongoing phase 3 trial investigating the efficacy of ^{177}Lu -PSMA-617 in patients with taxane-naïve mCRPC.

METHODS: Taxane-naïve patients with prostate-specific membrane antigen (PSMA)-positive mCRPC who had progressed once on prior ARPI were randomised 1:1 to open-label intravenous ^{177}Lu -PSMA-617 7.4 GBq (200 mCi) $\pm$ 10% once every 6 weeks for six cycles or change of ARPI (to abiraterone or enzalutamide, administered orally on a continuous basis as per product labelling). Crossover from ARPI change to ^{177}Lu -PSMA-617 was allowed after centrally confirmed radiographic progression. The primary endpoint was rPFS. Safety was a secondary endpoint.

FINDINGS: Overall, 468 patients were randomised to ^{177}Lu -PSMA-617 (n=234) or ARPI change (n=234). Baseline characteristics were generally well balanced between groups. Median number of ^{177}Lu -PSMA-617 cycles was 6 (interquartile range, 2). Of patients randomised to ARPI change, 134 patients (57%) crossed over to receive ^{177}Lu -PSMA-617. In the primary analysis, the hazard ratio (HR) for radiographic progression or death for ^{177}Lu -PSMA-617 vs ARPI change was 0.41 (95% confidence interval [CI], 0.29–0.56; $p < 0.0001$; median rPFS, 9.30 months [95% CI, 6.77–not estimable] vs 5.55 [4.04–5.95]). In an updated analysis at the third interim overall survival analysis, the HR for rPFS was 0.49 (95% CI, 0.39–0.61; median rPFS, 11.60 months [95% CI, 9.30–14.19] vs 5.59 [4.21–5.95]). The incidence of grade 3–5 adverse events was lower with ^{177}Lu -PSMA-617 vs ARPI change (81/227 [36%] vs 112/232 [48%]; grade 3–4, 80/227 [35%] vs 111/232 [48%]).

INTERPRETATION: ^{177}Lu -PSMA-617 prolonged rPFS relative to ARPI change, with a favourable safety profile.

FUNDING: Novartis; [ClinicalTrials.gov, NCT04689828](https://clinicaltrials.gov/ct2/show/study/NCT04689828).

Introduction

Mortality remains high among patients with metastatic castration-resistant prostate cancer (mCRPC) despite the availability of multiple life-prolonging therapy options.¹ Androgen receptor pathway inhibitors (ARPIs) have been the most common first-line systemic therapy for mCRPC. However, as ARPIs are increasingly initiated in earlier disease settings, the first-line therapy landscape for mCRPC is evolving.^{2–6} For patients who progress on a first-line ARPI, selection of the next appropriate therapy can be controversial. There is evidence that taxane-based chemotherapy is the most active therapy in this population.^{7,8} Based on this evidence, taxane therapy after initial progression on an ARPI is recommended by many clinical guidelines.^{9–11} Despite this, real-world data indicates that many patients and physicians select alternative therapies.^{2–5,12} Reasons for deferring or avoiding taxanes include patient and/or physician desire to avoid side effects that can negatively impact quality of life,^{13,14} an assessment that the therapy benefit does not outweigh negative side effects, or patient health status that renders taxanes medically inappropriate. As a

result, many patients receive alternative therapies.^{15–17} A change in ARPI is a common second-line therapy strategy among patients considered appropriate for delaying taxane-based chemotherapy.^{2–5,7,18–21} Alternatives include poly(ADP) ribose (PARP) inhibitors for patients with homologous recombination repair (HRR) gene alterations,^{22–24} sipuleucel-T for asymptomatic or minimally symptomatic disease,²⁵ pembrolizumab for patients with microsatellite instability-high or mismatch repair-deficient tumours,²⁶ and radium-223 for metastatic bone disease.²⁵ However, only a minority of patients receive such therapies. Therefore, there remains an unmet need for therapy options among patients who wish to defer taxane-based chemotherapy.

[¹⁷⁷Lu]Lu-PSMA-617 (¹⁷⁷Lu-PSMA-617; also known as lutetium vipivotide tetraxetan) is a prostate-specific membrane antigen (PSMA)-targeted radioligand therapy approved for PSMA-positive progressive mCRPC following both ARPI and taxane therapy.^{27–30} The VISION trial demonstrated the efficacy of ¹⁷⁷Lu-PSMA-617 in this post-taxane mCRPC setting, with significant prolongation of radiographic progression-free survival (rPFS) and overall survival following addition of ¹⁷⁷Lu-PSMA-617 to standard of care *vs* standard of care alone.^{31,32} In the phase 2 TheraP trial in the post-docetaxel mCRPC setting, ¹⁷⁷Lu-PSMA-617 was associated with improved intermediate endpoints such as PSA response *vs* cabazitaxel, with no difference in overall survival.^{33,34} PSMA is often overexpressed on prostate cancer cells but has restricted expression in most normal tissues, enabling ¹⁷⁷Lu-PSMA-617 to deliver β -particle radiation specifically to prostate cancer cells and the surrounding tumour microenvironment while minimising off-target effects.^{35,36}

PSMAfore is a phase 3 trial that was designed to compare ¹⁷⁷Lu-PSMA-617 with a change of ARPI in taxane-naïve patients with mCRPC who had progressed once on ARPI as last therapy, and were considered candidates for ARPI change. We report the primary endpoint of rPFS, the planned third interim analysis of the key secondary endpoint of overall survival, and several other secondary and exploratory endpoints.

Methods

TRIAL OVERSIGHT

PSMAfore ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT04689828) number, [NCT04689828](https://clinicaltrials.gov/ct2/show/study/NCT04689828)) was conducted in accordance with the Declaration of Helsinki principles and Good Clinical Practice guidelines. Independent ethical review boards at all sites approved the protocol (available at [thelancet.com](https://www.thelancet.com)). All patients provided written informed consent.

The trial was sponsored by Novartis and designed and overseen by a steering committee comprising the lead trial investigators and sponsor representatives. Data were analysed by the sponsor and interpreted in collaboration with the lead investigators. All authors had full access to the data, vouch for the fidelity of the trial to the protocol, and critically reviewed and approved the manuscript. An independent data monitoring committee assessed the progress of the trial, safety data, and critical efficacy variables.

PATIENTS

Eligible patients were adults with PSMA-positive mCRPC who were candidates for a change of ARPI following one instance of disease progression on abiraterone, enzalutamide, darolutamide, or apalutamide as their most recent anticancer therapy. Candidacy for change of ARPI was determined by the treating physician during the informed consent discussion and was based on the patient's clinical history. Patients were ineligible if they had received chemotherapy for castration-resistant or hormone-sensitive prostate cancer (six or fewer cycles of adjuvant or neoadjuvant taxane therapy ≥ 12 months ago was allowed), immunotherapy (except sipuleucel-T) or systemic/hemi-body radiotherapy (≤ 6 months ago), or were known to have actionable genomic alterations (e.g. *AR-V7* or *BRCA1/2* alterations) for which alternative standard therapies were available (e.g. PARP inhibitors). Eastern Co-operative Oncology Group (ECOG) performance status of 0 or 1 and adequate organ function were required.

The criteria for PSMA-positive mCRPC on centrally read baseline [^{68}Ga]Ga-PSMA-11 (^{68}Ga -PSMA-11) positron emission tomography-computed tomography (PET-CT) scans required patients to have at least one PSMA-positive (^{68}Ga -PSMA-11 uptake greater than liver parenchyma) metastatic lesion and no PSMA-negative lesions exceeding specified minimum dimensions. All lymph nodes measuring ≥ 25 mm in the short axis, bone metastases with a soft tissue component ≥ 10 mm in the longest diameter, solid organ metastases ≥ 10 mm in the longest diameter, and all intraprostatic lesions had to be ^{68}Ga -PSMA-11 PET positive. Read criteria differed from VISION only in the requirement for any intraprostatic lesions to be PSMA positive in PSMAfore.^{31,37} Full eligibility criteria are provided in the protocol (available at thelancet.com).

DESIGN AND INTERVENTIONS

This open-label, phase 3 trial was conducted at 74 sites (25 in North America and 49 in Europe). Patients were randomised 1:1 to receive either ^{177}Lu -PSMA-617 7.4 GBq (200 mCi) $\pm 10\%$ once every 6 weeks for six cycles, or a change of ARPI to abiraterone or enzalutamide as per treating investigator selection (Figure S1). Patients in the ARPI change group could crossover to ^{177}Lu -PSMA-617 upon meeting eligibility criteria, including radiographic disease progression confirmed by blinded independent central review (BICR) as per Prostate Cancer Working Group 3³⁸ (PCWG3)-modified Response Evaluation Criteria in Solid Tumours (RECIST) v1.1. Crossover was not allowed upon clinical progression in the absence of centrally confirmed radiographic progression.

Permitted and prohibited concomitant therapies are detailed in the Supplementary Methods.

Treatment continued until completion of six cycles of ^{177}Lu -PSMA-617, unacceptable toxicity, radiographic progression, requirement for a prohibited therapy, or withdrawal. Participants had CT or magnetic resonance imaging and $^{99\text{m}}\text{Tc}$ bone scans every 8 weeks for 24 weeks, then every 12 weeks, until radiographic progression per PCWG3-modified RECIST v1.1. Participants attended follow-up visits within 1 week of last treatment or radiographic progression, 1 month thereafter for safety, then every 12 weeks (Figure S1).

RANDOMISATION AND MASKING

Randomisation was performed using interactive response technology and was stratified according to previous ARPI use (castration-resistant *vs* hormone-sensitive prostate cancer), and by pain symptoms (asymptomatic or mildly symptomatic *vs* symptomatic; score of 0–3 *vs* >3 on item 3 of the screening Brief Pain Inventory - Short Form [BPI-SF]). Before randomisation, abiraterone or enzalutamide was chosen by the investigator as the change of ARPI; the choice was disregarded if participants were randomised to receive ¹⁷⁷Lu-PSMA-617. Treatment assignment was open to participants, investigator staff, personnel performing assessments, and the Novartis study team. However, no aggregate summaries by treatment group were performed by Novartis until the primary rPFS analysis. Independent central reviewers were blinded to randomised treatment assignment.

ENDPOINTS

The primary endpoint was rPFS, defined as time from randomisation to radiographic disease progression (by BICR as per PCWG3-modified³⁸ RECIST v1.1), or death. The key secondary endpoint was overall survival, defined as time from randomisation to death from any cause. Other secondary and exploratory endpoints are listed in Table S1. The following secondary endpoints were immature at the time of the third interim overall survival analysis and are not reported here: second rPFS (for patients who crossed over), time to soft tissue progression, and time to chemotherapy.

Efficacy was analysed in all randomised participants. Safety was assessed in all participants who received at least one dose of study treatment.

STATISTICAL ANALYSES

The primary analysis of rPFS was prespecified to occur after approximately 156 events in 450 patients, providing 95% power to detect a hazard ratio (HR) of 0.56 at an overall one-sided α of 0.025. The key secondary endpoint was to be tested if the primary endpoint was met in the primary analysis. A Lan–DeMets (O’Brien–Fleming) α spending function was used to control type I error across four-look testing of overall survival. Interim analyses of overall survival were prespecified. The primary intent of the interim analyses was to stop early for superior efficacy; there was no intent to assess futility at the interim analyses. The first interim overall survival analysis was to occur at the time of the primary analysis of rPFS. Second and third interim overall survival analyses were to occur at approximately 125 and 223 deaths, respectively. Final analysis of overall survival is to occur after approximately 297 deaths, providing 80% power to detect a hazard ratio of 0.72 (not considering the impact of crossover) at an overall one-sided α of 0.025. Updated analyses of rPFS could occur at the time of the second and third interim overall survival analyses.

Time-to-event endpoints were assessed using the log-rank test stratified by the randomisation factors. Hazard ratios and confidence intervals (CIs) were estimated using the stratified Cox proportional-hazards model. Medians and confidence intervals were estimated using the Kaplan–Meier method. P values were converted to two-sided for the primary endpoint. Analysis methods for other endpoints and further details are provided in the Supplementary Appendix.

ROLE OF THE FUNDING SOURCE

This study was funded by Novartis, which, in collaboration with the authors, was involved in the study design, collection, analysis, and interpretation of the data, and the writing and review of the manuscript.

Results

PATIENTS

Of 585 patients screened, 547 underwent ^{68}Ga -PSMA-11 PET-CT and 505/547 (92%) met the PSMA PET inclusion criteria (Figure 1). Overall, 468/585 patients (80%) met all the eligibility criteria and were randomised between June 15, 2021 and October 7, 2022 to receive either ^{177}Lu -PSMA-617 (n=234) or ARPI change (n=234). Protocol deviations are listed in Table S2. At third data cut-off, 134/234 patients (57%) randomised to the ARPI change group had crossed over to receive ^{177}Lu -PSMA-617; these represented 78% of patients who had radiographic progression by BICR.

The primary analysis, as well as the second and third interim overall survival analyses, were complete at the time of publication. The study is ongoing ahead of the final analysis of overall survival. The first data cut-off was October 2, 2022 for the primary analysis. The second data cut-off was June 21, 2023, and the third data cut-off was February 27, 2024 for the second and third interim overall survival analyses, respectively. All data presented in this manuscript, other than the primary analysis of rPFS, are based on the third data cut-off for the third interim overall survival analysis. Data corresponding to the second data cut-off have been previously presented.⁴² The median time from randomisation to first data cut-off was 7·26 months (interquartile range [IQR], 7·16) and to third data cut-off was 24·11 months (7·16).

Baseline patient characteristics were generally well balanced between groups (Table 1). Abiraterone was the prior ARPI received in 119/234 patients (51%) in the ^{177}Lu -PSMA-617 group and 130/234 (56%) in the ARPI change group. The proportion of patients with ECOG performance status of 0 was higher in the ^{177}Lu -PSMA-617 group (146/234 [62%]) than in the ARPI change group (116/234 [50%]).

After discontinuation of study treatment, for ^{177}Lu -PSMA-617 vs ARPI change, 110/234 patients (47%) vs 99/234 (42%) received chemotherapy, 44/234 (18%) vs 41/234 (18%) received radiotherapy, and 31/234 (13%) vs 5/234 (2%) received hormonal therapy. Other post-protocol anticancer therapies were received by less than 5% of patients in either group (Table S3).

EFFICACY

Primary endpoint: In the primary analysis (first data cut-off) after 166 rPFS events, the hazard ratio for radiographic progression or death with ^{177}Lu -PSMA-617 vs ARPI change was 0·41 (95% CI, 0·29–0·56; two-sided $p < 0·0001$); the median rPFS was 9·30 months (95% CI, 6·77–not estimable [NE]) vs 5·55 months (4·04–5·95) (Figure S2). The median follow-up time for rPFS in the full analysis set was 3·71 months (IQR, 3·91) in the

^{177}Lu -PSMA-617 group (median number of cycles among patients who received at least one dose of ^{177}Lu -PSMA-617, 3.0; IQR, 3.0 [Table S4]) and 2.35 months (3.75) in the ARPI change group.

At third data cut-off after 334 rPFS events, the updated hazard ratio for radiographic progression or death with ^{177}Lu -PSMA-617 vs ARPI change was 0.49 (95% CI, 0.39–0.61); the median rPFS was 11.60 months (95% CI, 9.30–14.19) vs 5.59 months (4.21–5.95) (Figure 2A). The median follow-up time for rPFS in all patients randomised was 8.77 months (IQR, 13.08) in the ^{177}Lu -PSMA-617 group (median number of cycles among patients who received at least one dose of ^{177}Lu -PSMA-617, 6.0; IQR, 2.0 [Table S4]) and 4.27 months (6.37) in the ARPI change group. In prespecified subgroup analyses of rPFS, hazard ratios were <1 for ^{177}Lu -PSMA-617 vs ARPI change in all subgroups (Figure S3).

Key secondary endpoint: At third data cut-off, there were 216 deaths, corresponding to 73% of the number of deaths prespecified in the statistical analysis plan for the final overall survival analysis. Death occurred in 104/234 (44%) patients in the ^{177}Lu -PSMA-617 group, including three deaths before treatment was started. Death occurred in 112/234 (48%) patients in the ARPI change group, including in 59/134 (44%) patients who crossed over. Most deaths were related to prostate cancer (^{177}Lu -PSMA-617 group, 81/104 [78%]; ARPI change group, 94/112 [84%]). In the intention-to-treat analysis, the hazard ratio for overall survival with ^{177}Lu -PSMA-617 vs ARPI change was 0.98 (95% CI, 0.75–1.28; $p < 0.44$); the median overall survival was 23.66 months (95% CI, 19.75–NE) vs 23.85 months (20.60–26.55) (Figure 2B). The median follow-up time for overall survival in all patients randomised was 17.94 months (IQR, 9.89) in the ^{177}Lu -PSMA-617 group and 18.45 months (8.21) in the ARPI change group.

Other efficacy endpoints: At third data cut-off, the proportion of patients with confirmed prostate-specific antigen (PSA) decline of at least 50% from baseline was higher with ^{177}Lu -PSMA-617 vs ARPI change (51% [95% CI, 43.8–57.5] vs 17% [12.6–22.9]) (Figure 3A). Median time to PSA progression was longer with ^{177}Lu -PSMA-617 vs ARPI change (10.64 months [95% CI, 8.57–12.68] vs 4.24 months [3.48–4.44]; HR, 0.40; 95% CI, 0.31–0.50) (Figure S4). For ^{177}Lu -PSMA-617 vs ARPI change, hazard ratios were 0.41 (95% CI, 0.26–0.63) for time to first symptomatic skeletal event (Table S5), 0.43 (0.35–0.52) for progression-free survival (Figure S5), and 0.86 (0.67–1.11) for progression-free survival on next line of therapy (which could include crossover therapy) (Figure S6).

The objective response rate for soft tissue disease, excluding bone lesion progression, among patients with measurable disease at baseline was higher with ^{177}Lu -PSMA-617 vs ARPI change (50% [95% CI, 38.0–62.0] vs 15% [7.9–25.7]), including 15/72 patients (21%) vs 3/72 (4%) achieving complete soft tissue response (Figure 3B and Table S6).

Patient-reported endpoints: At third data cut-off, median time to worsening in Functional Assessment of Cancer Therapy-Prostate (FACT-P) total score was longer with ^{177}Lu -PSMA-617 vs ARPI change (7.46 months [95% CI, 6.08–8.54] vs 4.27 months [3.45–4.50]; HR, 0.61 [95% CI, 0.50–0.75]) (Figure 2C), as was time to worsening in BPI-SF pain intensity (Figure 2D).

SAFETY

Exposure: At third data cut-off, patients in the ^{177}Lu -PSMA-617 group had received a median cumulative dose of 42.39 GBq (IQR, 14.27), and the median duration of exposure was 8.41 months (IQR, 2.96) (Table S4). In the ARPI change group, the median duration of exposure was 6.52 months (IQR, 7.77); 100 patients received abiraterone and 132 patients received enzalutamide.

Treatment-emergent adverse events: The incidence of treatment-emergent adverse events (TEAEs) of grade 3 or higher was lower with ^{177}Lu -PSMA-617 vs ARPI change (81/227 patients [36%] vs 112/232 [48%]) (Table 2). These TEAEs were considered treatment-related by the investigator in 27/227 patients (12%) in the ^{177}Lu -PSMA-617 group and in 29/232 (13%) in the ARPI change group (Table S7). Dry mouth, asthenia, nausea, and anaemia were the four most common TEAEs in the ^{177}Lu -PSMA-617 group. Most of these events were grade 1 or 2 (Table 2). Thrombocytopenia was observed in 17/227 patients (7%) in the ^{177}Lu -PSMA-617 group and 7/232 (3%) in the ARPI change group (Table S7). TEAEs leading to dose adjustment and treatment interruption were less frequent in the ^{177}Lu -PSMA-617 group than in the ARPI change group (Table 2). The proportion of patients who experienced TEAEs leading to treatment discontinuation or death was low and comparable between groups (Table 2). There were no drug-related TEAEs leading to death in ^{177}Lu -PSMA-617 group and one in the ARPI change group (cerebrovascular accident) (Table S7). Groupings of TEAEs according to safety topics of interest are shown in Table S8. The number of patients who experienced renal toxicity adverse events was similar in the ^{177}Lu -PSMA-617 (15/227 [7%]) and ARPI change (18/232 [8%]) groups (Table S8). Serious and grade 5 treatment-related adverse events are detailed in Supplementary Results. Adverse events relating to ^{68}Ga -PSMA-11 are detailed in Table S9.

Discussion

Taxane-naïve patients with mCRPC who have progressed on a first ARPI face an important therapy sequencing decision. Taxane-based chemotherapy has proven efficacy in relieving symptoms, delaying disease progression, and prolonging survival, but toxicity can negatively impact quality of life during treatment.^{8,13,14} Many patients forego taxanes in this clinical context in favour of switching to an alternative ARPI.^{2-5,12} This trial demonstrates that ^{177}Lu -PSMA-617 offers a more effective therapy option for controlling disease and maintaining quality of life than changing ARPI. The results of this study are complementary to those of the pivotal VISION study, which assessed ^{177}Lu -PSMA-617 in a post-taxane patient population.

Treatment with ^{177}Lu -PSMA-617 led to a clinically meaningful prolongation of rPFS compared with ARPI change in taxane-naïve patients with PSMA-positive mCRPC. ^{177}Lu -PSMA-617 had a favourable safety profile, with a lower incidence of grade 3–4 or serious adverse events despite a longer median duration of exposure than ARPI change. Other patient-relevant and clinically important endpoints also favoured ^{177}Lu -PSMA-617 over ARPI change, including time to symptomatic skeletal events, health-related quality of

life, and radiographic and biochemical response rates, as well as investigator-assessed progression-free survival and progression-free survival on next line of therapy.

The estimation of median rPFS was unstable at the time of the primary analysis owing to the short follow-up; median rPFS had stabilised by third data cut-off, showing a clinically meaningful difference between medians in favour of ¹⁷⁷Lu-PSMA-617.

Overall survival was the key secondary endpoint in this study. At the time of the third data cut-off, there was no difference in overall survival between the ¹⁷⁷Lu-PSMA-617 and ARPI change groups in the intention-to-treat analysis. The trial was designed to allow crossover from ARPI change to ¹⁷⁷Lu-PSMA-617 to offer access to ¹⁷⁷Lu-PSMA-617 in a patient-centric study design. The high crossover rate (134/234 patients [57%] randomised to ARPI change) may have confounded the analysis of overall survival. Methods for adjusting overall survival for the effect of crossover will be explored in the planned final overall survival analysis.

PSMAfore compared ¹⁷⁷Lu-PSMA-617 with a change in ARPI in patients who, in conjunction with their consulting physician, felt that it was appropriate to change ARPI and delay taxane-based chemotherapy. A change in ARPI has lesser clinical efficacy than taxanes,⁵ and it may not be suitable for all patients. However, real-world data indicates that many patients do not receive taxane-based chemotherapy after progression on a first ARPI and instead switch to a second ARPI.^{2–5,12} Here, median rPFS in the ARPI change group was 5.59 months at third data cut-off, which is longer than previously reported in several studies,^{23,44,45} and PSA response was consistent with published literature,^{21,23,46} supporting the use of ARPI change as a control treatment for the PSMAfore population. Only patients who were candidates for a change in ARPI were enrolled, and this was reflected in the baseline patient characteristics, which were consistent with patients having more indolent disease. For example, the proportion of patients with liver metastases in this trial was low (6% in the ¹⁷⁷Lu-PSMA-617 group and 3% in the ARPI change group), and was consistent with a phase 2 trial of enzalutamide and abiraterone sequencing in patients with newly diagnosed mCRPC.¹⁹ Patients could discontinue the trial at any time pre- or post-randomisation to receive other anticancer therapies, including chemotherapy; patients with clinical progression had the option to discontinue randomised treatment and receive chemotherapy, even in the absence of radiographic progression. At third data cut-off, less than half of patients had received subsequent chemotherapy. These observations combined with the multiple patient-centric benefits afforded with ¹⁷⁷Lu-PSMA-617 suggest that a change in drug class to ¹⁷⁷Lu-PSMA-617 could offer an acceptable alternative to change of ARPI for second-line therapy in mCRPC.^{2,18–21} PSMAfore did not have a bidirectional crossover design, so conclusions cannot be made about the optimal sequencing of change of ARPI vs second-line ¹⁷⁷Lu-PSMA-617.

PSMA PET imaging criteria were met in 92% of patients who underwent ⁶⁸Ga-PSMA-11 PET–CT scanning, while 87% of patients met the PSMA PET imaging criteria in VISION.³¹ Further analyses of PSMA PET imaging parameters are planned.

This study has several limitations. Firstly, it did not examine a higher risk population for whom taxane-based chemotherapy was the most appropriate next therapy after progression on an ARPI, therefore, the results cannot be generalised to this population. Phase 2 trials of ^{177}Lu -PSMA-617 in the pre-taxane mCRPC setting vs docetaxel (NCT04663997) and in the metastatic hormone-sensitive prostate cancer setting (NCT04343885) before docetaxel are ongoing. However, future larger studies using a chemotherapy comparator are warranted for this higher risk population. Secondly, PSMAfore did not assess the differential efficacy of ^{177}Lu -PSMA-617 in patients with or without pathogenic HRR gene alterations, which are associated with high tumour PSMA expression⁴⁷ and are recognised in mCRPC.^{48–51} Patients with known HRR gene alterations were ineligible. However, genotyping was not required during screening so it is not known what proportion of patients had HRR mutations, and the results cannot be generalised to patients with HRR gene alterations. Finally, double-blinding was not possible in this trial because of the different administration routes of the trial interventions and because of radiation protection regulations and ethical considerations. This could have biased the reporting of clinical and safety endpoints, particularly those which have an element of subjectivity, including health-related quality of life and adverse event reporting. The lack of double-blinding may also have influenced the ongoing management of patients.

In conclusion, PSMAfore met the primary endpoint, demonstrating significant prolongation of time to radiographic progression or death with ^{177}Lu -PSMA-617 compared with ARPI change in taxane-naïve patients with mCRPC and disease progression on prior ARPI. In addition, ^{177}Lu -PSMA-617 was more effective than change of ARPI in prolonging the time to biochemical progression, symptomatic skeletal events, deterioration in health-related quality of life, and worsening of pain, with a favourable safety profile. Overall survival was similar with ^{177}Lu -PSMA-617 and ARPI change; whether or not the high rate of crossover confounded the analysis of overall survival will be explored at the time of the final analysis. These results support ^{177}Lu -PSMA-617 as a treatment option for patients with mCRPC who have initially progressed on an ARPI and who are being considered for a change of ARPI as their next therapy.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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DATA SHARING

Novartis is committed to sharing patient-level data and supporting clinical documents from eligible studies with qualified external researchers. Requests are reviewed and approved by an independent review panel on the basis of scientific merit. All data provided are anonymised to respect the privacy of patients who participated in the trial, in line with applicable laws and regulations. Data availability for this trial is according to the criteria and process described on www.clinicalstudydatarequest.com.

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RESEARCH IN CONTEXT

Evidence before this study

[¹⁷⁷Lu]Lu-PSMA-617 (¹⁷⁷Lu-PSMA-617) is a prostate-specific membrane antigen (PSMA)-targeted radioligand therapy for treatment of patients with PSMA-positive metastatic castration-resistant prostate cancer (mCRPC) who have previously received androgen receptor pathway inhibitor (ARPI) and taxane therapy. We used the term ‘(metastatic castration-resistant prostate cancer) AND (lutetium) AND randomized NOT review[pt]’ to search PubMed for English language publications from January 1, 2014 to May 13, 2024. We identified three completed and published randomised controlled phase 2 clinical trials and the pivotal phase 3 VISION trial. One of the phase 2 trials ([NCT03392428](#)) reported higher PSA response and fewer grade 3 or 4 adverse events with ¹⁷⁷Lu-PSMA-617 compared with cabazitaxel in 200 post-docetaxel patients. Another of the phase 2 trials ([NCT04419402](#)) reported PSA progression-free survival was longer with ¹⁷⁷Lu-PSMA-617 plus enzalutamide vs enzalutamide alone as first-line therapy in 162 patients with mCRPC. A third phase 2 trial (CTRI/2019/12/022282) compared ¹⁷⁷Lu-PSMA-617 with docetaxel prechemotherapy but the sample size was very small. The pivotal VISION trial ([NCT03511664](#)) showed that addition of ¹⁷⁷Lu-PSMA-617 to standard of care improved radiographic progression-free survival and overall survival compared with standard of care alone in patients with progressive, extensively pre-treated mCRPC.

Added value of this study

To our knowledge, PSMAfore is the second phase 3 clinical trial of ¹⁷⁷Lu-PSMA-617 in patients with prostate cancer and is the first phase 3 trial to assess the efficacy and safety of ¹⁷⁷Lu-PSMA-617 in the taxane-naïve mCRPC population. The results show that ¹⁷⁷Lu-PSMA-617 is an effective alternative to a change of ARPI for delaying disease progression and deterioration of quality of life in patients with mCRPC who have initially progressed on an ARPI and for whom it was considered appropriate to delay taxane-based chemotherapy. The results of PSMAfore complement those of VISION, which assessed ¹⁷⁷Lu-PSMA-617 in the post-taxane mCRPC population.

Implications of all the available evidence

The findings of PSMAfore extend the high-quality body of evidence for the efficacy and safety of ¹⁷⁷Lu-PSMA-617 in patients with mCRPC. The previous phase 3 VISION trial and randomised controlled phase 2 trials provided evidence for the efficacy of ¹⁷⁷Lu-PSMA-617 in patients previously treated with both ARPIs and taxane-based chemotherapy. PSMAfore now provides evidence that ¹⁷⁷Lu-PSMA-617 is a more effective therapy option with a better safety profile than change of ARPI for achieving disease control, demonstrated by prolonged radiographic progression-free survival, in patients in whom it is considered appropriate to change ARPI and delay taxane-based chemotherapy.

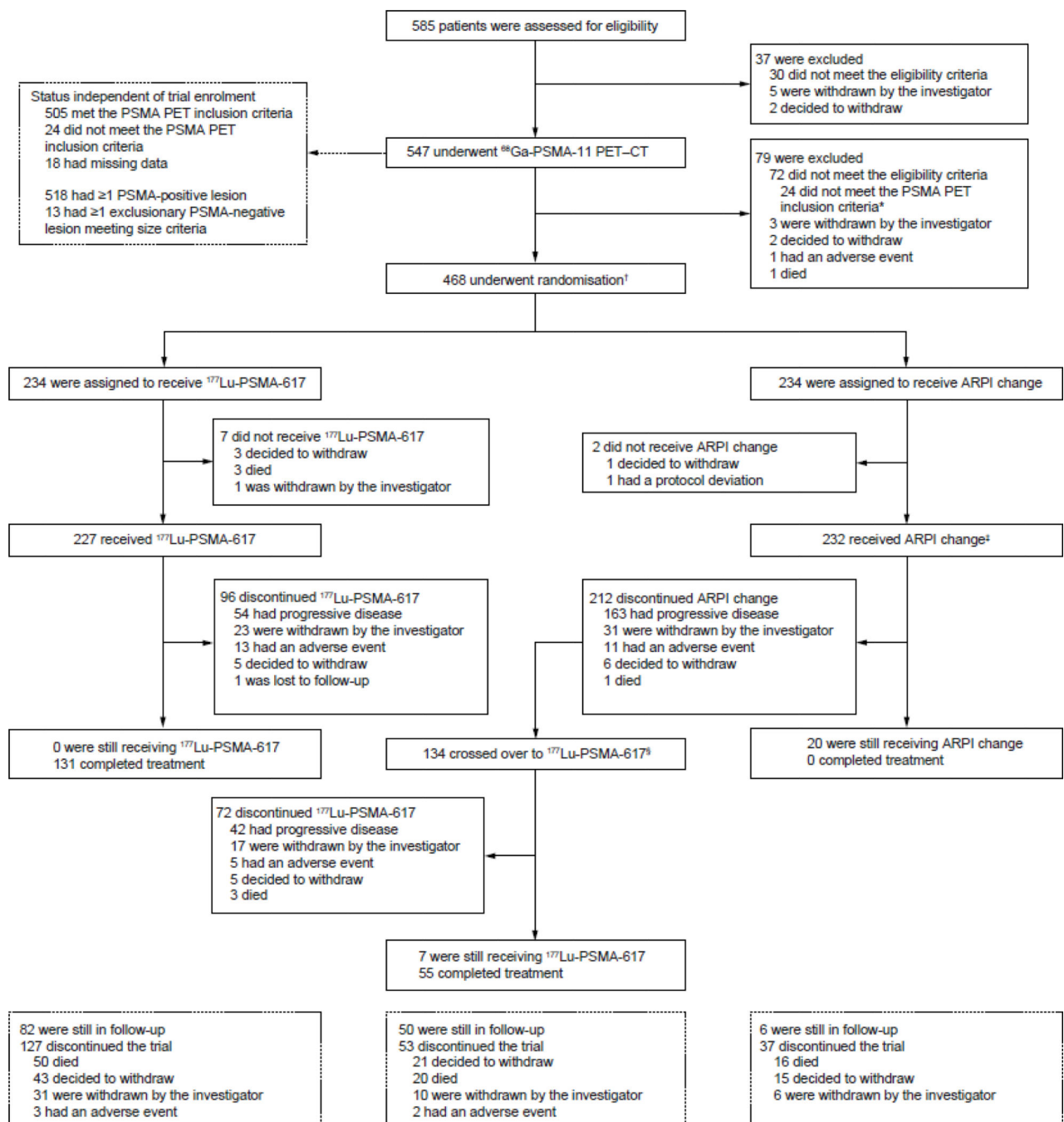


Figure 1: Screening, randomisation, and follow-up at the time of third interim overall survival analysis

* 37 met PSMA PET imaging criteria but were excluded because they did not meet the other eligibility criteria or for other reasons.

† One additional patient who underwent randomisation and was assigned to ^{177}Lu -PSMA-617 is not shown because they had a protocol deviation of informed consent not signed and were excluded from all analyses. One patient was randomised after the data cut-off for the primary analysis of the primary endpoint.

‡ Abiraterone, n=100; enzalutamide, n=132.

§ Patients eligible to crossover to ^{177}Lu -PSMA-617 had discontinued ARPI change with radiographic disease progression by blinded independent central review as per RECIST v1.1. ARPI=androgen receptor pathway inhibitor; CT=computed tomography; PET=positron-emission tomography; PSMA=prostate-specific membrane antigen.

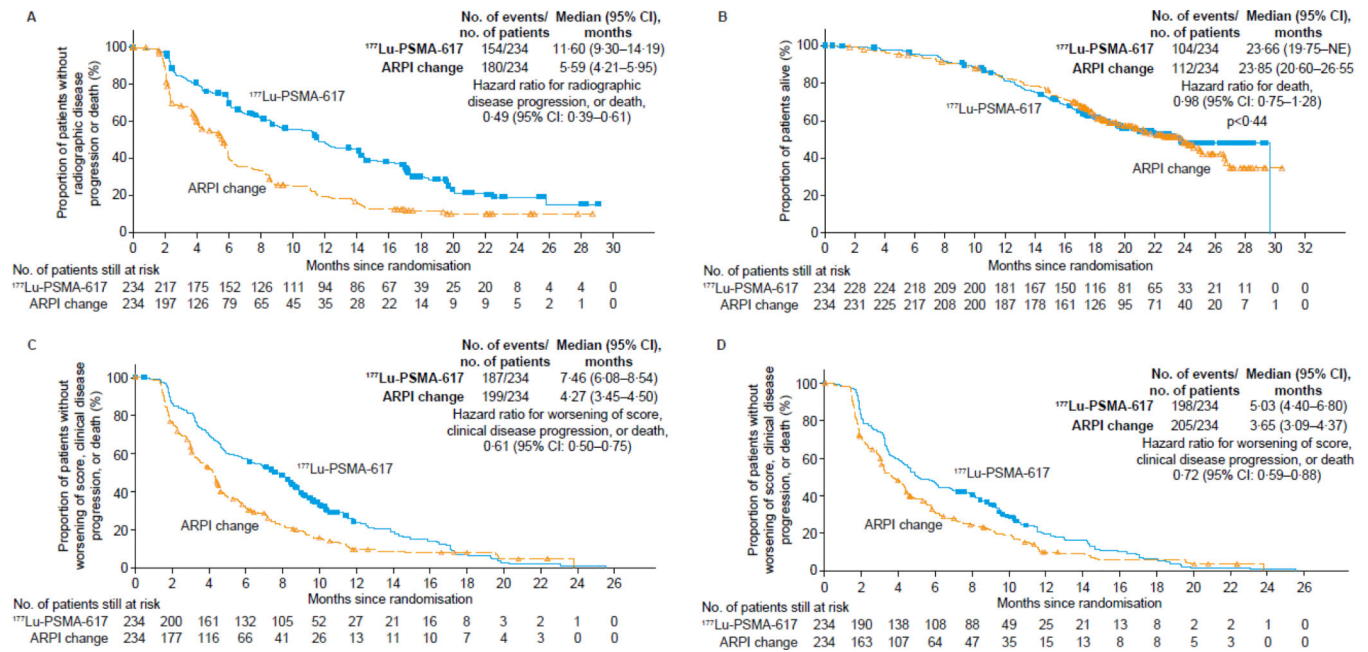


Figure 2: Radiographic progression-free survival (primary efficacy endpoint), overall survival (key secondary efficacy endpoint), health-related quality of life (secondary efficacy endpoint), and patient-reported pain (exploratory efficacy endpoint)

Panel A shows the updated analysis of radiographic progression-free survival at the time of the third interim overall survival analysis. The primary analysis of radiographic progression-free survival is shown in Figure S2. Radiographic progression-free survival was defined as the time to radiographic disease progression by blinded independent central review as per the Prostate Cancer Clinical Trials Working Group 3 criteria, or death.

Panel B shows the third interim intention-to-treat analysis of overall survival. Note: three patients died before receiving ^{177}Lu -PSMA-617.

Panel C shows the time to worsening of 10 points or greater in FACT-P total score^{52,53}, disease progression, or death at the time of the third interim overall survival analysis.

Panel D shows the time to worsening of 2 points or greater in BPI-SF pain intensity scale^{54,55}, clinical disease progression, or death at the time of the third interim overall survival analysis.

Schedule of investigations is detailed in Figure S1.

ARPI=androgen receptor pathway inhibitor; BPI-SF=Brief Pain Inventory-Short Form; CI=confidence interval; FACT-P=Functional Assessment of Cancer Therapy-Prostate; NE=not estimable; PSMA=prostate-specific membrane antigen.

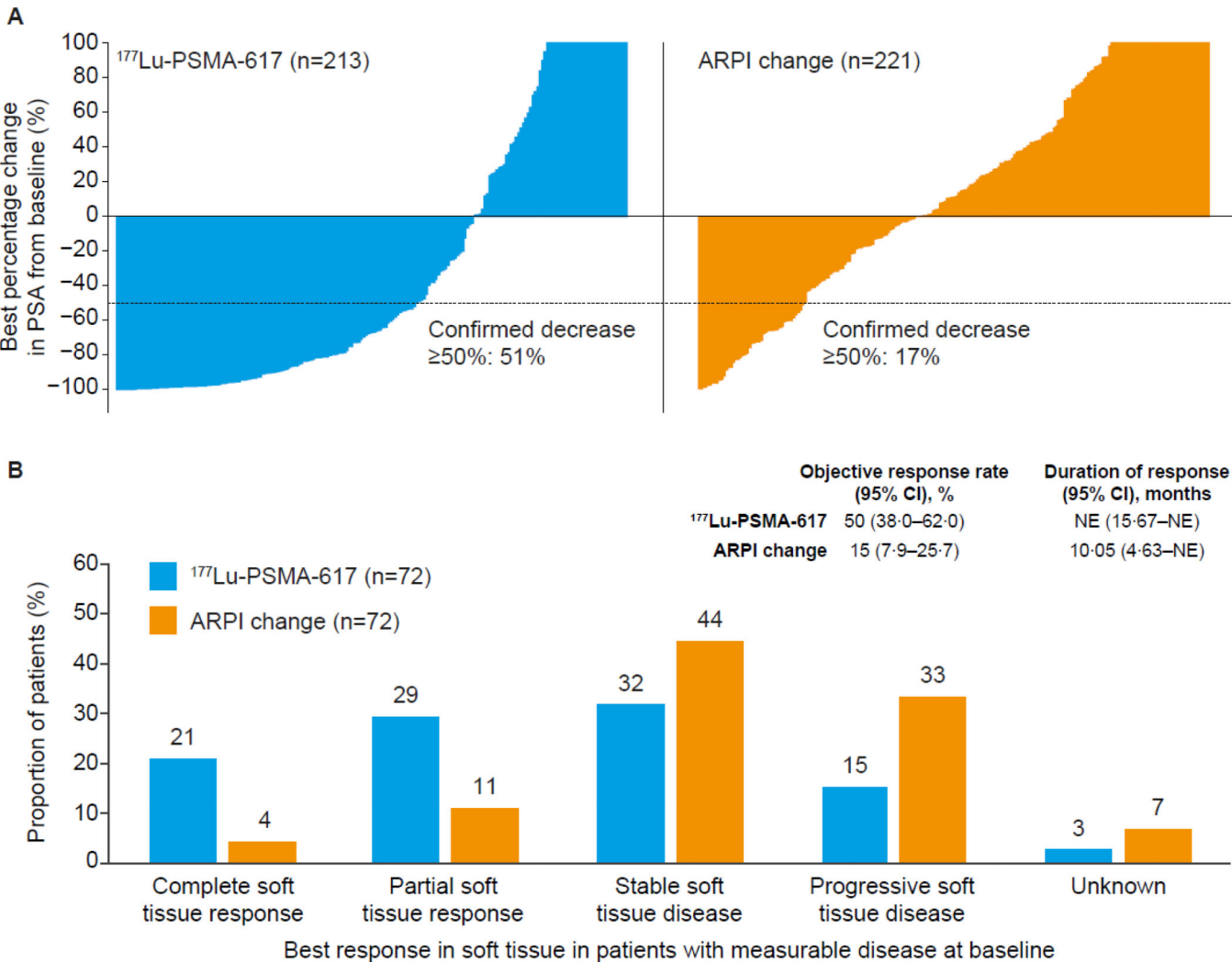


Figure 3: Prostate-specific antigen responses (secondary efficacy endpoint) and soft tissue responses* (exploratory efficacy endpoint)

Panel A shows PSA responses. Data are from patients with available PSA measurements at the time of the third interim overall survival analysis; increases greater than 100% are truncated to 100%. Assessments were at 3, 6, and 12 months after randomisation.

Panel B shows objective response rate. Data are from a subset of patients with measurable disease at baseline at the time of the third interim overall survival analysis; objective response rate is the sum of complete soft tissue response and partial soft tissue response; duration of response is estimated using the Kaplan–Meier method in patients with objective response as the duration from first documented best overall response to documented disease progression or death.

* Assessment of soft tissue response was based on RECIST v1.1 but excluded bone lesion progression. ARPI=androgen receptor pathway inhibitor; CI=confidence interval; NE=not estimable; PSA=prostate-specific antigen; PSMA=prostate-specific membrane antigen; RECIST=Response Evaluation Criteria in Solid Tumours.

Table 1:
Patient baseline characteristics *

	¹⁷⁷ Lu-PSMA-617(n=234)	ARPI change (n=234)
Region – no. (%)		
Europe	187 (80)	182 (78)
North America	47 (20)	52 (22)
Age		
Median (IQR) – years	71·0 (12·0)	72·0 (10·0)
<65 years – no. (%)	50 (21)	44 (19)
≥65 years – no. (%)	184 (79)	190 (81)
OMB category – no. (%)		
White	212 (91)	214 (91)
Black or African-American	7 (3)	5 (2)
Asian	1 (0)	2 (1)
Unknown	14 (6)	13 (6)
ECOG performance status – no. (%)		
0	146 (62)	116 (50)
1	86 (37)	114 (49)
Missing	2 (1)	4 (2)
Site of disease – no. (%)		
Lung	21 (9)	23 (10)
Liver	13 (6)	7 (3)
Lymph node	76 (32)	73 (31)
Bone	205 (88)	203 (87)
RECIST v1·1 soft tissue lesions – no. (%)		
Target only	24 (10)	25 (11)
Non-target only	39 (17)	39 (17)
Target and non-target	58 (25)	62 (26)
None	113 (48)	108 (46)
Haemoglobin, median (IQR) – g/L [†]	128·0 (17·0)	129·0 (15·0)
Alkaline phosphatase, median (IQR) – IU/L [‡]	100·0 (86·0)	103·5 (66·0)
Lactate dehydrogenase, median (IQR) – IU/L [‡]	197·0 (65·0)	196·5 (57·0)
Prostate-specific antigen, median (IQR) – µg/L [§]	18·4 (44·7)	14·9 (32·4)
Prostate cancer history		
Time since diagnosis, median (IQR) – years	5·45 (7·80)	6·80 (7·70)
Gleason score – no. (%)		
3	0	2 (1)
4	1 (0)	3 (1)
5	4 (2)	1 (0)
6	15 (6)	11 (5)
7	69 (29)	95 (41)
8	46 (20)	46 (20)
9	76 (32)	55 (24)
10	15 (6)	9 (4)

	¹⁷⁷ Lu-PSMA-617(n=234)	ARPI change (n=234)
Missing	8 (3)	12 (5)
Previous surgery (non-biopsy) – no. (%)	82 (35)	96 (41)
Previous radiotherapy – no. (%)	150 (64)	172 (74)
Previous chemotherapy setting – no. (%)		
Neoadjuvant	1 (0)	0
Adjuvant	1 (0)	1 (0)
Metastatic	4 (2) ^{//}	5 (2) ^{//}
Previous anticancer regimens – no. (%)		
1	43 (18)	46 (20)
2	119 (51)	118 (50)
3	54 (23)	53 (23)
4	13 (6)	13 (6)
5	4 (2)	3 (1)
6	0	1 (0)
10	1 (0)	0
Setting at last anticancer regimen – no. (%)		
Adjuvant	2 (1)	4 (2)
Locally advanced/premetastatic	7 (3)	10 (4)
Metastatic	225 (96)	220 (94)
Previous androgen receptor pathway inhibitor – no. (%)		
Abiraterone	119 (51)	130 (56) [¶]
Enzalutamide	94 (40)	83 (35)
Other ^{**}	21 (9)	21 (9)
Previous androgen receptor pathway inhibitor setting – no. (%) ^{††}	190 (81)	190 (81)
Castration-resistant prostate cancer	44 (19)	44 (19)
Hormone-sensitive prostate cancer		
Pain symptoms – no. (%) ^{††}		
Asymptomatic or mildly symptomatic ^{‡‡}	165 (71)	166 (71)
Symptomatic ^{§§}	69 (29)	68 (29)

* Shown are data for all patients who were randomised. Percentages may not total 100 because of rounding.

[†]¹⁷⁷Lu-PSMA-617, n=227; ARPI change, n=232.

[‡]¹⁷⁷Lu-PSMA-617, n=223; ARPI change, n=230.

[§]¹⁷⁷Lu-PSMA-617, n=230; ARPI change, n=229.

^{//} Protocol deviations.

[¶] Two patients discontinued abiraterone because of toxicity and subsequently received darolutamide prerandomisation.

^{**} Apalutamide or darolutamide.

^{††} Randomisation stratification factor.

^{##}Score of 0–3 on BPI-SF item 3 (worst pain intensity).

^{\$\$}Score of >3 on BPI-SF item 3 (worst pain intensity).

ARPI=androgen receptor pathway inhibitor; BPI-SF=Brief Pain Inventory-Short Form; ECOG=Eastern Co-operative Oncology Group; IQR=interquartile range; OMB=Office of Management and Budget; PSMA=prostate-specific membrane antigen.

Table 2:
Overview of treatment-emergent adverse events*

Patients with treatment-emergent adverse events – no. (%)	¹⁷⁷ Lu-PSMA-617 (n=227)		ARPI change (n=232)	
	Any grade	Grade ≥3	Any grade	Grade ≥3
Any	224 (99)	81 (36) [†]	226 (97)	112 (48) [†]
Occurring in >10% of patients				
Dry mouth	131 (58) [‡]	2 (1)	6 (3) [‡]	0
Asthenia	74 (33)	2 (1)	67 (29)	8 (3)
Nausea	72 (32)	0	27 (12)	1 (0)
Anaemia	61 (27)	14 (6)	44 (19)	16 (7)
Fatigue	53 (23)	1 (0)	59 (25)	4 (2)
Constipation	50 (22)	1 (0)	33 (14)	0
Decreased appetite	49 (22)	0	43 (19)	1 (0)
Arthralgia	45 (20)	0	54 (23)	1 (0)
Diarrhoea	38 (17)	0	21 (9)	1 (0)
COVID-19	36 (16)	1 (0)	27 (12)	1 (0)
Back pain	31 (14)	3 (1)	46 (20)	6 (3)
Vomiting	26 (11)	0	11 (5)	0
Oedema peripheral	19 (8)	0	28 (12)	0
Pain in extremity	17 (7)	1 (0)	25 (11)	1 (0)
Weight loss	15 (7)	1 (0)	32 (14)	3 (1)
Leading to adjustment of dose	8 (4)	3 (1)	36 (16)	7 (3)
Leading to interruption of treatment	28 (12)	13 (6)	45 (19)	20 (9)
Leading to discontinuation of treatment	13 (6)	8 (4)	12 (5)	9 (4)
Leading to death [§]	4 (2)	4 (2)	5 (2)	5 (2)

* Shown are data for patients who underwent randomisation and received at least one dose of their assigned treatment; seven patients who were randomly allocated to the ¹⁷⁷Lu-PSMA-617 group did not receive treatment.

[†] Grade 3–4: n=80 (35%) in the ¹⁷⁷Lu-PSMA-617 group; n=111 (48%) in the ARPI change group.

[‡] Grade 2 dry mouth: n=19 in ¹⁷⁷Lu-PSMA-617 group; n=0 in ARPI change group.

[§] In the ¹⁷⁷Lu-PSMA-617 group: COVID-19 pneumonia, cardiac arrest, intestinal ischaemia, and sepsis (all n=1; all considered not related to study drug by the investigator); in the ARPI change group: cardiac arrest, coma, dyspnoea, multiple organ dysfunction syndrome (all n=1; all considered not related to study drug by the investigator), and cerebrovascular accident (n=1; considered related to study drug by the investigator).

ARPI=androgen receptor pathway inhibitor; PSMA=prostate-specific membrane antigen.